

RED DOG DOCK, AK, USA

WMO: 997704

Lat: 67.575N

Lon: 164.067W

Elev: 10

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -22.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 50.6	(k) 17.9	(l) 44.2	(m) 21.9	(n) 6.4	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 66.6	(d) N/A	(e) 63.1	(f) N/A	(g) 60.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 8.0	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 42.5	(b) 35.5	(c) 30.5	(e) -31.4	(f) 75.5	(g) 6.2	(h) 6.4	(i) -35.9	(j) 80.1	(k) -39.5	(l) 83.9	(m) -43.0	(n) 87.4	(o) -47.6	(p) 92.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	3.4	6.9	6.0	17.5	34.1	43.4	51.9	50.7	43.0	29.3	14.5	8.0
	DBStd	20.88	17.47	16.04	12.95	11.51	8.97	7.28	6.20	5.91	7.42	7.04	10.80	13.50
	HDD50	9067	1446	1208	1365	974	496	226	49	62	231	642	1066	1302
	HDD65	14304	1911	1628	1830	1424	959	648	409	444	661	1107	1516	1767
	CDD50	241	0	0	0	0	3	28	107	83	20	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	13	0	0	0	0	0	2	10	1	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	11.0	12.3	15.4	11.9	9.3	9.6	7.2	9.2	9.1	10.7	11.8	11.9	14.3
Precipitation	PrecAvg	17.8	1.3	1.4	1.0	0.8	0.7	0.8	1.9	2.8	2.0	2.4	1.5	1.2
	PrecMax	24.8	2.7	4.4	5.0	2.2	1.8	1.5	3.9	6.1	4.4	4.6	4.6	3.1
	PrecMin	12.8	0.3	0.2	0.2	0.1	0.0	0.4	0.5	0.6	0.5	0.9	0.2	0.4
	PrecStd	3.1	0.6	1.0	0.9	0.6	0.5	0.3	0.8	1.3	0.8	0.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	38.0	35.2	44.2	60.0	68.7	74.6	70.4	62.2	47.0	39.2	37.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	35.2	34.6	32.5	40.3	54.8	62.4	67.5	65.3	57.9	43.2	36.2	34.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	31.3	31.8	29.2	37.5	50.5	58.3	63.8	62.0	55.2	41.2	33.2	30.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	27.3	29.3	26.0	34.7	46.2	54.2	60.6	59.3	53.1	39.1	30.0	26.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.9	13.9	15.1	12.6	11.9	9.6	9.3	10.6	8.8	9.3	11.0
		MCDBR	12.3	11.0	12.4	15.1	18.6	17.8	15.7	12.9	10.7	7.5	7.5	10.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.173	0.237	0.259	0.326	0.365	0.345	0.356	0.362	0.311	0.252	0.171	N/A
	taud		1.898	2.082	2.217	2.057	2.085	2.399	2.444	2.422	2.505	2.304	1.952	N/A
	Ebn at Noon		49	184	253	257	259	272	265	249	237	183	46	0
	Edh at Noon		3	16	24	38	42	32	30	27	19	12	3	0
All-Sky Solar Radiation	RadAvg		16	180	740	1440	1816	1949	1518	1109	671	231	36	0
	RadStd		2	23	56	75	172	169	159	139	73	27	5	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page